Does Doubling the Halfway Split Predict the Finish? A Simple Linear Regression of the 2023 Boston Marathon

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The code used to download, prepare, and analyze the data is available at https://github.com/pujithabobbili/simple-regression-project.

Abstract

Doubling a runner's halfway time is a common way to project a marathon finish, but it assumes that the two halves take equal time. Using official results for 26,526 finishers of the 2023 Boston Marathon (Fay et al. 2023), I fitted a simple linear regression of net finishing time on half-marathon split, both in minutes, and tested whether the slope differs from 2, the value implied by equal halves. The estimated slope was 2.218 (95% confidence interval 2.212 to 2.224), so each extra minute at halfway is associated with about 1.22 extra minutes in the second half, and this held with robust standard errors and a log-log model. As a result, doubling underestimated the finish for about 89% of runners, by about 5 minutes on average at a 1:30 split and about 18 minutes at a 2:30 split. The halfway split predicts the average finish well, but an individual runner's finish remains uncertain by roughly 19 minutes either way, so the fitted equation is best used as an estimate of the mean.

1 Introduction

If a runner reaches the halfway point of a marathon in 1 hour 45 minutes, doubling that time gives a projected finish of 3 hours 30 minutes. However, it works exactly only when the runner completes the second half in another 1 hour 45 minutes. Any slowdown after halfway makes the projection too optimistic.

Running the second half more slowly than the first is called a positive split. Previous studies have found that positive splits are common in marathons. March et al. (2011) and Santos-Lozano et al. (2014) reported differences in pacing across runners of different abilities, and March et al. found that slower and less experienced runners tended to slow more. Runners slow down for several reasons, such as using up stored glycogen and building up muscle damage. Earlier research also found that men slowed more than women on average (Deaner et al. 2015), that warmer conditions were associated with larger pacing changes, especially among slower runners (Ely et al. 2008), and that some recreational runners have severe late-race slowdowns (Smyth 2021). These studies compare groups of runners, but they do not give one finishing-time equation that a runner could use at halfway.

In this paper, I ask two questions. First, how closely does the halfway split predict net finishing time in the 2023 Boston Marathon? Second, does the fitted slope differ from 2, the value expected if the two halves take equal time? To answer them, I fitted a simple linear regression of net finishing time on half-marathon split for the 26,526 finishers with a recorded split. Regression software usually tests whether a slope is zero, but that is not the main question here: the halfway time is already part of the finishing time, so a strong positive association is expected. Testing against 2 is more useful because it compares the fitted relationship with the doubling rule.

The estimated slope is 2.218. Therefore, an extra minute at halfway is associated with about 2.22 extra minutes at the finish, including about 1.22 extra minutes in the second half. This matters in practice: doubling understates the finish, and the shortfall grows for slower runners. The conventional prediction intervals extend about 19 minutes on either side of the fitted value, so the equation estimates the average finish better than any one runner's. Section 2 describes the data, Section 3 explains the model and its checks, Section 4 presents the results, and Section 5 discusses their limits and possible extensions.

2 Data

2.1 Source and observational units

I used the official results of the Boston Marathon held on April 17, 2023. Fay et al. (2023) compiled the file from results published by the Boston Athletic Association (2023) and made it available through the SCORE Network Sports Data Repository. Each row represents one finisher. The full file has 26,598 finishers, so it covers the finishing field rather than a random sample of it. Runners who started but did not finish are not included.

The race times come from electronic timing chips. Mats along the course record when a runner passes, and the organizer publishes the results after the race. They do not show what happened between checkpoints.

Since the complete finishing field is observed, the fitted line describes the runners in this race directly. The confidence intervals and tests require a broader interpretation. I use them as model-based measures of uncertainty for similar runners racing under conditions like these, rather than as uncertainty from sampling the 2023 finishing field.

Boston also has a selective entry process. Most entrants meet a qualifying-time standard at an earlier marathon, while other runners enter through charity programs. Its runners are therefore not a random sample of everyone who runs marathons. This affects how widely the results can be applied.

2.2 Variables

The source file contains 15 columns with runner information, finishing places, and race times. I used five columns and calculated two additional variables. The model itself uses only net finishing time and the half-marathon split. Sex and age group describe the field; gun time helps explain the choice of response.

Net finishing time is the elapsed time between crossing the starting mat and crossing the finishing mat. I converted it from seconds to minutes. This is different from gun time, which begins at the official start signal. Boston uses waves and starting corrals, so a runner may cross the starting mat several minutes after the signal. In this file, gun time exceeds net time by a median of 2.5 minutes and by as much as 27.1 minutes. Net time is the appropriate response for studying the runner's pace because it excludes this waiting time.

Half-marathon split is the net elapsed time at the halfway mat, also converted to minutes. This is the explanatory variable. From the split and finishing time, I calculated the second-half time (net finishing time minus the split) and the split ratio (second-half time divided by the split). A split ratio above 1 means a slower second half. A ratio below 1 means a faster second half, and a ratio of 1 means that both halves took the same time.

Sex is recorded as M or W in the source file. Women make up 42.9% of the analysis data. Age is given in ten groups, from 18–39 to 80+, rather than as an exact value. The largest group is 18–39, with 38.2% of finishers; the remaining 61.8% are aged 40 or over.

The times are recorded to about one second, a small amount compared with the 19.3-minute standard deviation of halfway splits. I therefore treated timing precision as negligible for this analysis. A more important issue is that finishing time includes the halfway split. The two variables share part of the same measurement, so their strong association needs to be interpreted with that structure in mind.

2.3 Missing data and field characteristics

There are 72 finishers without a recorded half-marathon split, or 0.27% of the full field. A missed chip reading is one possible explanation, although the results file does not establish why a split is missing. Since these runners lack the predictor needed for the model, I excluded them. This leaves 26,526 runners. No other variable used in the analysis has missing values, and there are no duplicate runners.

The excluded runners have a mean finishing time of 224.4 minutes, compared with 222.5 minutes for those retained. Their sex and age-group distributions are also similar to the rest of the field. The exclusions are therefore unlikely to have a large effect.

Table 1 summarizes the analysis variables. Halfway times range from 1:02:19 to 3:27:15, and net finishing times range from 2:05:54 to 7:20:31. The means are above the medians for all three time variables, reflecting the long upper tails. The median split ratio is 1.058: a runner at the median of this ratio took about 5.8% longer for the second half.

Table 1: Summary statistics for the 26,526 finishers with a recorded halfway split. Times are in minutes. The split ratio has no units. All three times are right skewed, with means above their medians, and the median split ratio of 1.058 means that the typical finisher ran the second half about 6% slower than the first.

Variable	Mean	SD	Min	Q1	Median	Q3	Max
Half-marathon split	106.9	19.3	62.3	92.5	103.5	116.9	207.2
Net finishing time	222.5	43.7	125.9	191.1	213.9	244.4	440.5
Second-half time	115.6	25.3	63.6	97.7	110.1	127.9	248.4
Split ratio	1.078	0.086	0.526	1.022	1.058	1.113	2.524

The 11,384 women had a median finish of 3:46:04, while the 15,142 men had a median finish of 3:20:56. Their pacing summaries were much closer. Both groups had a median split ratio of 1.058. A slower second half was recorded for 88.7% of women and 89.3% of men.

Comparisons with other marathons are also limited because all runners faced the same course and race day. Earlier work links temperature to pacing, particularly for slower runners (Ely et al. 2008), but this file does not provide variation across race days that would allow me to estimate a separate weather effect.

Figure 1 shows the distributions of halfway time, finishing time, and the split ratio. Both time distributions have long right tails. Most split ratios are just above 1, consistent with a modest slowdown, but a smaller number are much larger. Overall, 89.0% of finishers ran a positive split.

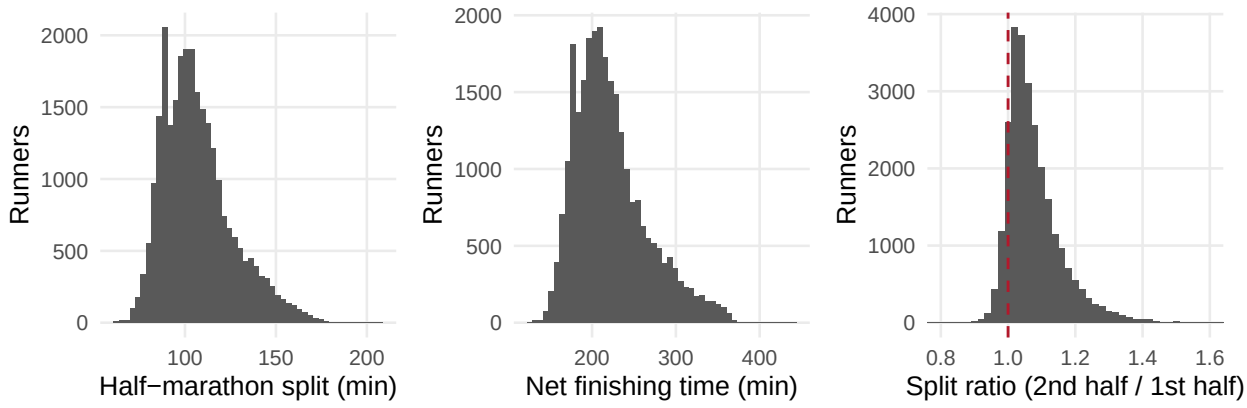


Figure 1: Distributions of halfway split, net finishing time, and split ratio. The dashed line in the right panel marks equal times for the two halves. Its horizontal axis is limited to 0.8–1.6; 0.15% of runners fall outside that range. The largest split ratio is 2.524. Both times have long right tails, and most split ratios are just above 1, so most runners slowed in the second half.

Figure 2 plots finishing time against halfway time. Each point represents a runner. The dashed line is $y = 2x$, the doubling rule. Points above it are positive splits. The main group of points follows a roughly straight pattern and sits mostly above the reference line. The gap also becomes larger toward the longer halfway times. This is the pattern the regression measures.

2.4 Additional and alternative data

Several additional data sources could help explain the pacing pattern. The Boston Athletic Association publishes 5 km splits, which could show where a runner began to slow. Results from other years and other marathons would allow comparisons across courses and conditions. Hourly weather records from the National Oceanic and Atmospheric Administration could be matched to the period when runners were on the course. Qualifying times would provide information about recent performance before race day.

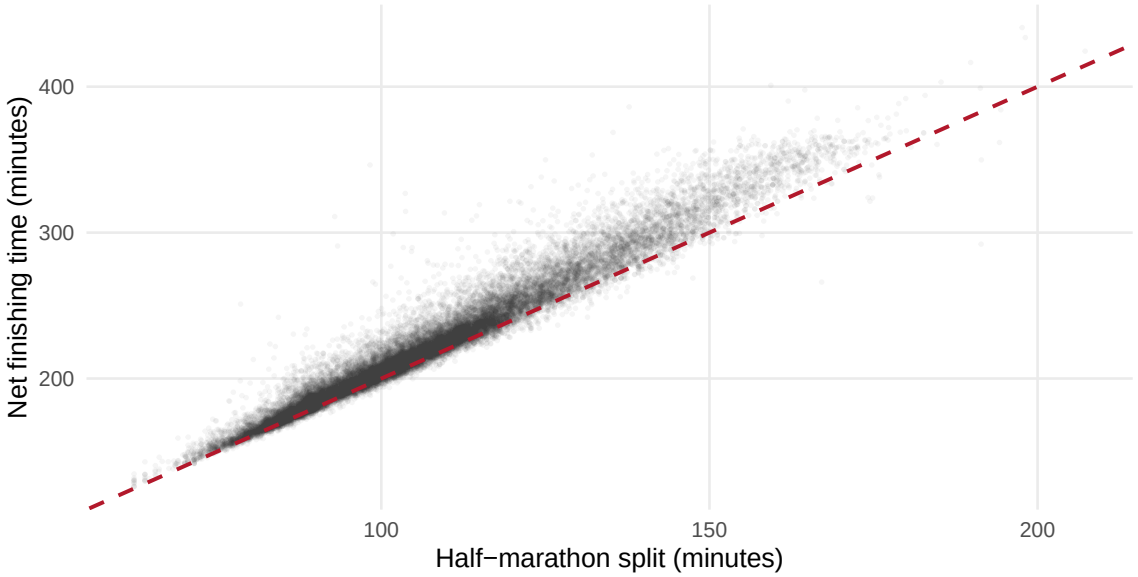


Figure 2: Net finishing time against halfway split before fitting the regression. Transparent points make areas with many runners appear darker. The dashed line represents equal times for the two halves, $y = 2x$. Most points lie above the line, and the gap widens for longer splits, which suggests a slope above 2.

3 Methods

3.1 The model

For runner i, let Y_i be net finishing time and x_i be the half-marathon split, both in minutes. There are $n = 26,526$ runners. The initial simple linear regression model is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2).$$

The intercept, β_0, is the model's expected finishing time at a halfway split of zero. That split is impossible in a race, so the intercept is useful for positioning the line rather than for describing an actual runner. The slope, β_1, is the expected difference in finishing time associated with a one-minute difference in halfway time. The error, ε_i, is the difference between runner i's finishing time and the conditional mean specified by the model. The errors are assumed to be independent and normally distributed with mean 0 and constant variance σ^2.

I estimated the coefficients using ordinary least squares. It chooses the line that minimizes

$$\sum_{i=1}^{n}(Y_i - \beta_0 - \beta_1 x_i)^2.$$

These are squared vertical distances from the observed points to the line. Once the model is fitted, each residual is the runner's observed finishing time minus their fitted finishing time. The residual standard deviation, $\hat{\sigma}$, summarizes the scale of these errors in minutes.

For a halfway split of x_0, the predicted finishing time is

$$\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0.$$

There are two different uncertainties to consider. A confidence interval for the mean finish concerns the average finishing time of runners with that split. A prediction interval concerns one runner's finishing time. The latter must allow for individual differences around the line as well as uncertainty in the estimated line, so it is much wider.

I used one predictor to keep the question focused on what can be estimated from a halfway time alone.

3.2 The even-pacing benchmark

If both halves took the same time for every runner, finishing time would satisfy $Y_i = 2x_i$. This corresponds to $\beta_0 = 0$ and $\beta_1 = 2$. I tested the slope against that benchmark using

$$H_0 : \beta_1 = 2 \qquad \text{versus} \qquad H_1 : \beta_1 \neq 2.$$

The conventional test statistic is

$$t = \frac{\hat{\beta}_1 - 2}{SE(\hat{\beta}_1)},$$

where $SE(\hat{\beta}_1)$ is the standard error of the estimated slope, with $n - 2 = 26,524$ degrees of freedom under the initial model assumptions. This tests the slope component of the equal-pacing relationship. A slope of 2 alone would not establish equal pacing because a nonzero intercept could still produce a systematic difference between the halves.

A slope above 2 means that a one-minute increase in halfway time is associated with more than two extra minutes in total finishing time. Since one of those minutes is already in the first half, more than one extra minute is associated with the second half.

A high R^2 needs care here, because the response contains the predictor as one of its parts. The estimated slope's distance from 2 and the spread of the residuals are more useful for the questions asked here.

3.3 Assumptions and diagnostics

I examined the main regression assumptions with plots and diagnostic summaries.

Linearity. The model assumes that mean finishing time changes at a constant rate as halfway time increases. I checked the scatterplot and a plot of residuals against fitted values for curvature. The main pattern is approximately straight, and the residual plot does not show a strong curved trend. A straight line is therefore a reasonable summary of the average relationship over the observed range.

Independence. The model treats the errors of different runners as unrelated. This is difficult to justify fully in a race. Runners may follow the same pacing group, run together, or respond similarly to course conditions. Dependence of this kind can make usual standard errors too small. The HC1 check below addresses unequal variance, not dependence between runners.

Constant variance. The initial model assumes the same error variance at every halfway time. The residual plot shows that this assumption does not hold well: the band becomes wider as fitted times increase. The

residual standard deviation is 5.9 minutes in the fastest tenth of the halfway-split distribution and 14.1 minutes in the slowest tenth. That is about a 2.4-fold increase. The Breusch–Pagan test also rejects constant variance, with $BP = 637.3$ on 1 degree of freedom and $p < 0.001$ (Breusch and Pagan 1979; Zeileis and Hothorn 2002).

Normality. The model assumes normally distributed errors. A histogram of the residuals is centered near zero, but it is not symmetric: it is right skewed, with a long upper tail of runners who finished much later than predicted, and both tails are heavier than a normal curve would give. With this many runners, the slope test is still reliable, but prediction intervals based on a normal error distribution remain approximations.

I also examined Cook's distance to assess how strongly individual observations affect the fitted line. The largest value is 0.060, below the commonly used reference value of 1. I kept runners with unusual pacing because they are real race results.

Figure 3 brings these checks together. Its main concern is the increasing spread of the errors, rather than obvious curvature. The bottom panel compares that spread on the original and log-log scales. Logging the times reduces the increase across the field, but it does not eliminate it.

3.4 Model selection and validation

I considered a model in minutes and a model using the natural logarithms of both times. I chose the original scale for the main analysis because its coefficients are easier to connect to the doubling rule. The slope can be compared directly with 2, and the residual standard deviation is expressed in minutes. I used the log-log fit to check whether the pacing conclusion depended on this choice of scale. The two models have different response variables, so comparing their R^2 values directly would not be a suitable way to choose between them.

Unequal variance affects the uncertainty calculations for the original model. The conventional standard errors assume a common variance and may be misleading here. A prediction interval using one residual standard deviation also cannot represent both the narrow spread among faster runners and the wider spread among slower ones.

I first recalculated the slope's standard error with a heteroskedasticity-consistent HC1 estimator (Zeileis 2004). This allows observations to contribute different amounts of error variance to the uncertainty estimate. It changes the standard error and confidence interval without changing the fitted line. As noted above, it does not account for runners' errors being correlated.

The second check was the log-log model,

$$\log(Y_i) = \gamma_0 + \gamma_1 \log(x_i) + \eta_i.$$

Here, γ_0 is the intercept, η_i is the error term, and γ_1 is an elasticity. A 1% increase in halfway time is associated with approximately a γ_1% increase in finishing time. Under equal pacing,

$$\log(Y_i) = \log(2) + \log(x_i),$$

so the slope benchmark is 1 and the intercept benchmark is $\log(2)$. Logging the variables compresses their long upper tails and can reduce the change in error spread. In this analysis it improves variance stability, although some unequal spread remains.

3.5 Possible extensions to the model

Adding sex and age group, together with interactions with the halfway split, would allow separate slopes for different groups. The similar overall split ratios for men and women do not tell us what those adjusted slopes would be.

A quadratic term or a spline could allow the average relationship to bend. The diagnostic plot does not show strong curvature, so I did not make this the primary model.

Weighted least squares could account for different amounts of error variance across halfway times, provided that suitable weights were estimated. Quantile regression could estimate selected finishing-time percentiles rather than only the mean. This would be useful for constructing prediction limits that vary with the split.

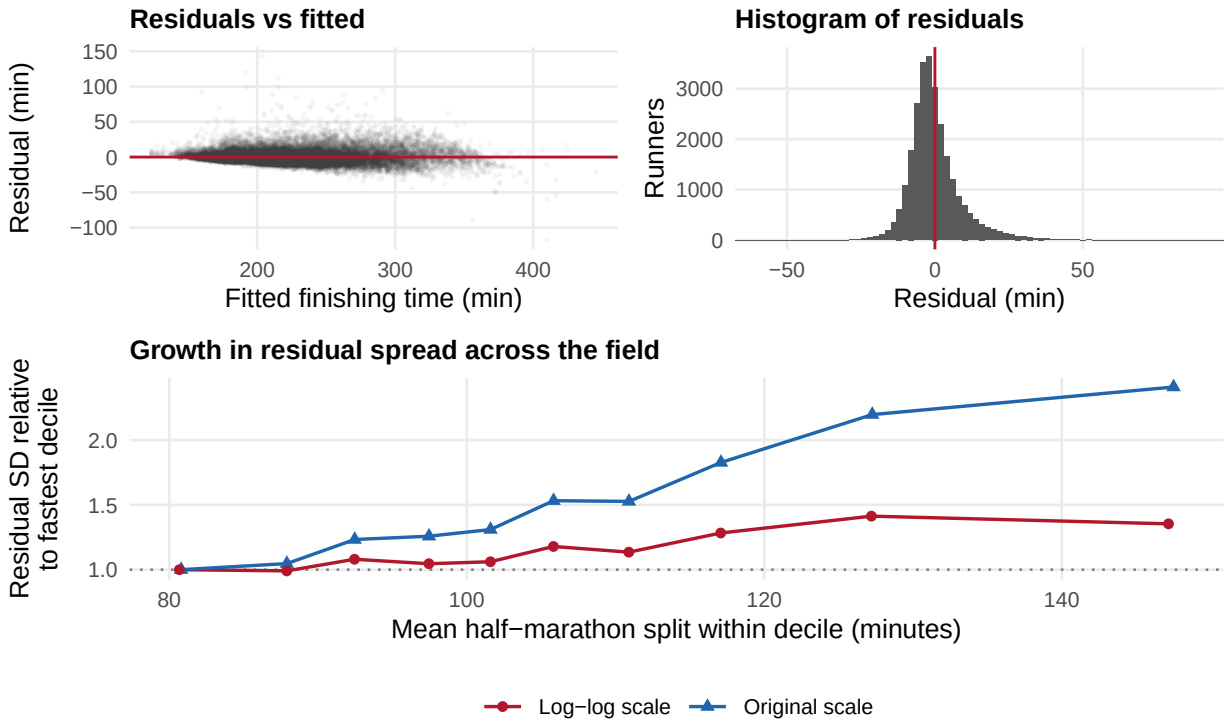


Figure 3: Regression diagnostics. The residual plot shows increasing spread at longer fitted times. The histogram of residuals is centered near zero but is right skewed, with a long upper tail; its horizontal axis is limited to −60 to 90 minutes. The bottom panel gives residual standard deviations within tenths of the halfway-split distribution, relative to the fastest tenth. Spread increases about 2.4-fold on the original scale and 1.4-fold on the log-log scale.

3.6 Software

The analysis was carried out in R version 4.6.1 (R Core Team 2026). I used `dplyr` for data handling (Wickham et al. 2023) and `readr` to read the data (Wickham et al. 2024). Figures were made with `ggplot2` (Wickham 2016) and combined using `patchwork` (Pedersen 2024). Robust standard errors came from `sandwich` (Zeileis 2004; Zeileis et al. 2020), and regression diagnostic tests used `lmtest` (Zeileis and Hothorn 2002). The report used Quarto (Allaire et al. 2022), `knitr` (Xie 2023), and `kableExtra` (Zhu 2024). The regressions were fitted with `lm` in R's `stats` package.

4 Results

4.1 The fitted model

The fitted line, with coefficients rounded for display, is

$$\widehat{\text{finish}} = -14.53 + 2.218 \times \text{half-marathon split}.$$

Both times are in minutes. Table 2 gives the coefficient estimates, conventional standard errors, and confidence intervals.

Table 2: Ordinary least squares estimates for net finishing time regressed on halfway split. The confidence intervals are conventional 95% intervals. The HC1 interval for the slope is 2.210–2.225. The slope of 2.218 is above the equal-pacing value of 2, and both intervals exclude 2.

Term	Estimate	Standard error	CI low	CI high
Intercept	-14.526	0.327	-15.167	-13.886
Half-marathon split	2.218	0.00301	2.212	2.224

The slope is above the equal-pacing benchmark of 2. Comparing it with that value gives $t = 72.4$ on 26,524 degrees of freedom, with $p < 0.001$. The conventional 95% confidence interval, 2.212–2.224, is entirely above 2. For runners whose halfway times differ by one minute, the model estimates a difference of about 2.22 minutes at the finish. Of that difference, one minute is in the first half and about 1.22 minutes is in the second half.

The negative intercept should not be interpreted as an actual finishing time. A zero-minute halfway split is impossible and is far outside the observed range of 62.3–207.2 minutes.

The model has $R^2 = 0.953$, and the correlation between halfway and finishing times is 0.976. These values show a close relationship in the observed field, but part of that relationship comes from finishing time containing halfway time. The residual standard deviation is 9.43 minutes.

Figure 4 adds the fitted line to the scatterplot. It crosses the doubling line at a halfway time of about 1:06:41. Only 30 runners reached halfway sooner than that. For all of the remaining field, the model predicts a finish later than simple doubling would predict. The difference grows with halfway time.

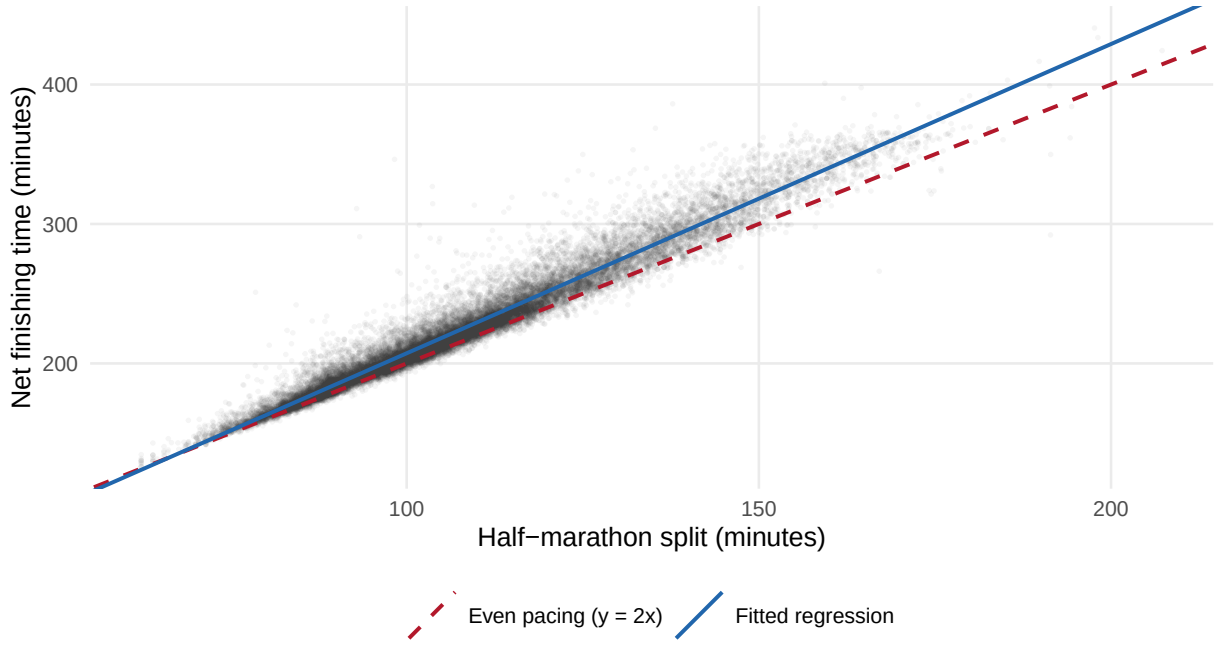


Figure 4: The fitted regression line and the doubling rule. The solid line is $\widehat{\text{finish}} = -14.53 + 2.218 \times \text{split}$; the dashed line is $\text{finish} = 2 \times \text{split}$. They cross near a 1:07 halfway time. Above that split, the fitted projection is later than the doubling projection.

4.2 What the model means for a runner

Table 3 gives projected finishing times for five halfway splits. It compares the doubling rule with the fitted model and includes approximate 95% prediction intervals for individual runners. The last column is the difference between the model projection and the doubling projection.

Table 3: Finishing-time projections from five halfway splits. Prediction intervals use the original model's common residual standard deviation and are approximate because error spread changes across the field and the residuals have heavy tails. The model adds more time to the doubled split as the halfway time grows, from about 5 to 18 minutes, and each prediction interval is about 37 minutes wide.

Halfway	Doubling	Model	95% prediction interval	Difference
1:30:00	3:00:00	3:05:05	2:46:35 to 3:23:34	5.1 min
1:45:00	3:30:00	3:38:21	3:19:51 to 3:56:50	8.3 min
2:00:00	4:00:00	4:11:37	3:53:07 to 4:30:06	11.6 min
2:15:00	4:30:00	4:44:53	4:26:23 to 5:03:22	14.9 min
2:30:00	5:00:00	5:18:09	4:59:39 to 5:36:38	18.1 min

At a 1:30 halfway time, the model adds about 5 minutes to the doubled split. At 2:00, it adds nearly 12 minutes. At 2:30, it adds about 18 minutes. For the median halfway split of 1:43:33, the difference is about 8 minutes. Differences of around a quarter of an hour or more occur near the slowest tenth of the field by halfway time. The doubling rule is therefore much closer to the fitted mean for faster runners than for slower runners.

One rough mental calculation is to double the halfway split and add about one minute for every five minutes beyond a split of 1:07. This approximates the fitted correction, since

$$\widehat{\text{finish}} - 2x = -14.53 + 0.218x.$$

The rule of thumb uses an increase of 0.2 rather than 0.218 minutes per minute, so it is only an approximation.

The prediction intervals in the table are about 37 minutes wide. For example, the model projection for a 1:45 split is 3:38:21, but the conventional interval extends from 3:19:51 to 3:56:50. The large field makes the mean relationship precise; it does not remove the variation in what individual runners do after halfway.

4.3 Robustness of the pacing conclusion

Table 4 compares the main model with the two checks. HC1 increases the slope standard error from about 0.0030 to 0.0039, roughly a 1.3-fold increase. The resulting interval is 2.210–2.225, and the test against 2 gives $t = 56.3$. Thus, allowing unequal error variance still leaves the estimated slope well above the equal-pacing benchmark.

The log-log slope is 1.058, compared with its benchmark of 1, with $t = 42.5$. On that scale, a 1% longer halfway time is associated with approximately a 1.058% longer total finishing time. The direction of the pacing result agrees with the original model. The change in residual spread across tenths is also smaller: about 1.4-fold on the log-log scale instead of 2.4-fold on the original scale. However, this is a reduction in unequal spread, not its complete removal.

Table 4: Checks of the slope conclusion. The benchmark is 2 on the original scale and 1 on the log-log scale. Slopes and standard errors are rounded for display; the t statistics were computed from unrounded values. In all three versions, the slope is well above its benchmark.

Model	Slope	Standard error	Benchmark	t
Original scale, conventional SE	2.218	0.0030	2	72.4
Original scale, robust HC1 SE	2.218	0.0039	2	56.3
Log-log scale, conventional SE	1.058	0.0014	1	42.5

These checks support the slope result, but they do not correct the original-scale prediction intervals in Table 3. Those intervals have nearly the same width at every split because they use one residual standard deviation. The observed spread is smaller among faster runners and larger among slower runners, so a single width is likely to be too wide at the faster end and too narrow at the slower end.

For a rough comparison, multiplying the fastest tenth's residual standard deviation of 5.9 minutes by 1.96 gives about 12 minutes. Doing the same with the slowest tenth's 14.1 minutes gives about 28 minutes. Both differ considerably from the common margin of about 18.5 minutes used in the table.

5 Discussion

For the 2023 Boston Marathon finishers analyzed here, the halfway split and finishing time have a strong, approximately linear relationship. The fitted slope is 2.218, above the value of 2 associated with equal halves. About 89% of runners actually ran a slower second half. Taken together, these results show why doubling the halfway time is usually optimistic for this field.

The gap between doubling and the fitted line grows for slower runners, which agrees with earlier findings that pacing varies with runner ability (March et al. 2011; Santos-Lozano et al. 2014). The present analysis provides an equation for this race's complete observed finishing field rather than only comparisons between broad ability groups.

However, a precise slope does not make a precise prediction for every runner. The overall residual standard deviation is 9.43 minutes, and the conventional individual prediction intervals extend roughly 19 minutes on either side of the fitted value. Their width also hides the change in error spread across halfway times. A projected finish should be treated as an estimate of the mean at that split, with substantial individual variation still possible.

The missing nonfinishers are an important limitation. Their absence means that this analysis describes only runners who completed the race. So the results do not show how much all starters slowed down. The selective Boston field also limits generalization. A different course, field, or race day could produce a different slope and a different amount of prediction error.

The assumptions create further limits. Error variance increases across the field, and the residuals have heavy tails. HC1 standard errors leave the main slope conclusion unchanged, but they do not repair the individual prediction intervals. These checks also do not account for runners who race together. With only one event, I cannot separate weather and course conditions from the relationship measured on that day.

This is an observational, descriptive model. It does not show that making the first half faster will cause a runner to finish sooner. A runner changing their own opening pace is not the same as comparing two runners with different halfway times.

Several follow-up analyses would be useful. Sex and age-group interactions could test whether the split-to-finish relationship differs across groups, as the earlier pacing literature suggests (Deaner et al. 2015). Multiple years and weather measurements could assess stability across race days and investigate temperature effects (Ely et al. 2008). The organizers' 5 km splits would help locate the onset of a slowdown instead of combining all second-half behavior into one time, which would be relevant to the late-race collapse studied by Smyth (2021).

Quantile regression could also give prediction limits that change with halfway time. For now, the fitted equation is a useful summary of the 2023 Boston finishers, with the clearest result being that the gap between simple doubling and the fitted mean becomes larger as halfway time increases.

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